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How Does the “Default Mode” Network Contribute to Semantic Cognition?

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Abstract

This review examines whether and how the “default mode” network (DMN) contributes to semantic processing. We review evidence implicating the DMN in the processing of individual word meanings and in sentence- and discourse-level semantics. Next, we argue that the areas comprising the DMN contribute to semantic processing by coordinating and integrating the simultaneous activity of local neuronal ensembles across multiple unimodal and multimodal cortical regions, creating a transient, global neuronal ensemble. The resulting ensemble implements an integrated simulation of phenomenological experience – that is, an embodied situation model – constructed from various modalities of experiential memory traces. These situation models, we argue, are necessary not only for semantic processing but also for aspects of cognition that are not traditionally considered semantic. Although many aspects of this proposal remain provisional, we believe it provides new insights into the relationships between semantic and non-semantic cognition and into the functions of the DMN.

1. Introduction

The close topographical resemblance between the “default mode” network (DMN) and the network implicated in semantic cognition raises the question of how activity in the DMN relates to semantic processing. This paper reviews evidence pointing to a functional role of the DMN in semantic processing and argues for an embodied model of semantic representation in which this network plays an essential role. We share the view of many authors that “meaning” depends on reference to the external world, and therefore on information learned from experience (Harnad, 1990; Searle, 1980). That is, learning the meaning of a new word consists of associating a generalization of a particular set of experiences to a word form. During this process, simple elements of experience such as shapes, colors, motor actions, affective states, and spatial relationships, to mention a few,

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are stored as local neuronal ensembles in cortical areas specialized in extracting each kind of information from the input provided by sensory organs and the motor apparatus, sometimes called “low level” or “unimodal” areas (Damasio, 1989; Pulvermüller, 1999). Word meaning is later retrieved – that is, activated in working memory – when these disparate memory traces are activated in synchrony and temporarily become entwined into a large-scale, distributed neuronal ensemble. The synchronous activation of experiential memory traces is coordinated by a set of high-level multimodal cortical hubs, which act as the functional spine of this dynamic, multi-regional network. In the following, we argue that this coordination and integration of synchronous activity is the main role played by the DMN, and that this role is required not only for the retrieval of linguistic meaning but for any task that requires the integration of multiple kinds of experiential information into a coherent internal model of the world. We use the term “experiential” to refer to any information encoded in the brain that originates from experience. It includes, for example, information directly extracted from a particular input channel, such as smell, color, sound, retinotopic motion, and nociceptive pain, as well as more elaborate kinds of information, generated from more than one input channel over the course of development, such as flavor, depth, egocentric spatial location, 3-dimensional shape and other physical properties of objects, spatial, temporal and causal relations, emotional states, and many other aspects of experience. Contrary to symbolic representations, which are arbitrarily related to the sensory-motor and affective properties of the entities represented, experiential representations are analogically related to the information encoded in sensory-motor and limbic areas (Harnad, 1990).

Perhaps the most fundamental fact about the functional organization of the cerebral cortex is that it contains areas dedicated to motor control, vision, hearing, somatosensation, taste, and smell, which are commonly referred to as modality-specific or unimodal areas. Some of these cortical regions receive the vast majority of their input from a single kind of sensory input, while others are primarily involved in generating motor output; damage to one of these areas causes modality-specific sensory or motor deficits (Mesulam, 1998). While these areas specialize in different aspects of the interface between the brain and the external environment, limbic/paralimbic structures such as the orbitofrontal cortex, insula, anterior cingulate gyrus, and amygdala are densely interconnected with the hypothalamus and brainstem homeostatic regions, providing information about the internal state of the organism (Craig, 2002). This modality of input to the brain, known as interoception, constitutes the basis for affective aspects of experience, such as emotional valence (Lindquist, Satpute, Wager, Weber, & Barrett, 2016; Seth, 2013). An equally fundamental observation is that most of the cortex is not dedicated to a particular sensory, motor, or interoceptive modality. Terms that are often used to refer to these areas include “association”, “transmodal”, “multimodal”, and “heteromodal” cortex, reflecting the fact that they receive and combine information originating from different input channels¹, although little is known about their computational properties. The pattern

¹Even though most of their output is directed at the primary motor cortex or at efferent motor neurons, motor areas also send copies of their output to primary somatosensory cortex, striatum, and cerebellum so it can be integrated with sensory information to generate internal models of the motor apparatus (Kawato, 1999; Mink, 1996; Shadmehr, Smith, & Krakauer, 2010). This reafferent input is known as an “efference copy” of the motor command.

of interconnectivity among these areas seems to follow a hierarchical organization, with primary sensory and motor areas on one end (typically thought of as the bottom of a vertical hierarchy), and areas that integrate information from different lower-level areas sitting at progressively higher levels, depending on the diversity of the information sources that are integrated (Margulies, Ghosh, Goulas, Falkiewicz, Huntenburg, et al., 2016; Mesulam, 1994; Sepulcre, Sabuncu, Yeo, Liu, & Johnson, 2012). In this paper, we used the term “multimodal” to denote areas that encode information originating from more than one input modality, qualifying them as relatively higher or lower in the hierarchy when necessary.

The term “default mode” network refers to a set of multimodal regions of the cerebral cortex that include, at a minimum, the posterior medial parietal cortex (posterior cingulate gyrus [pCinG] and precuneus), the medial and dorsal aspects of the superior frontal gyrus (SFG) and adjacent rostral cingulate gyrus, often referred to collectively as dorsomedial prefrontal cortex (DMPFC), and the angular gyrus (AG), all bilaterally (Buckner, Andrews-Hanna, & Schacter, 2008; Fox, Snyder, Vincent, Corbetta, Van Essen, et al., 2005). The term is more often applied to a larger set of regions that, in addition to the aforementioned ones, also includes portions of the lateral temporal cortex (particularly the middle temporal gyrus [MTG] and superior temporal sulcus [STS]), a portion of the parahippocampal gyrus (PHG), the retrosplenial cortex (RSC), ventral portions of the medial prefrontal cortex (VMPFC), lateral portions of the SFG and adjacent posterior middle frontal gyrus (pMFG), ventral portions of the inferior frontal gyrus (IFG), and the posterior supramarginal gyrus (SMG) (Figure 1A; Fox et al., 2005; Spreng, Mar, & Kim, 2009; Yeo, Krienen, Sepulcre, Sabuncu, Lashkari, et al., 2011).

As is well known, the term was coined by Raichle and colleagues (Raichle, MacLeod, Snyder, Powers, Gusnard, et al., 2001) in a classic paper addressing the phenomenon of task-induced “deactivation” observed in functional neuroimaging experiments, in which a set of cortical regions appeared to be consistently more active during periods of passive or “resting” states than during tasks that required attention to incoming sensory stimuli, such as visual search or perceptual discrimination (Shulman, Fiez, Corbetta, Buckner, Miezin, et al., 1997). Their chief concern at the time was to establish a physiological definition of “baseline” activity in the human brain, which they conceptualized as a state in which “equilibrium exists between the local metabolic requirements necessary to sustain a long-term modal level of neural activity and the level of blood flow in a particular region”. This definition was further operationalized as the state in which there is maximum uniformity across brain regions in the oxygen extraction fraction (OEF), which is the proportion of oxygen extracted from the local blood supply. Although not explicitly discussed, the authors’ intuition appears to have been that spatial uniformity indicates homeostasis, an unproven but not unreasonable assumption. Raichle et al. proposed that the eyes-open “resting” state (i.e., when the participant is lying motionless but awake in the scanner bed and not engaged in any explicit task) best satisfied this definition, but it is important to emphasize that their conception of “baseline” does not at all imply absent or even low levels of neural activity. Indeed, Raichle et al. explicitly stated that regions showing task-induced deactivation “may be tonically active in the baseline state”, and they focused much of their discussion on possible functions served by these tonically active areas. Whether or not the awake “resting” state should be considered a baseline, or indeed whether defining any

particular state as a baseline has any value for functional imaging studies, is beyond the scope of this paper. The important message we take from Raichle et al. is that neural activity in those cortical areas that typically show task-induced deactivation is driven by processes that are relatively more active when individuals are not given explicit tasks or disturbed by external stimuli.

Functional connectivity studies have indicated that, despite their broadly distributed locations across the brain, the various areas that make up the DMN typically work together as a functional unit. Unlike other regions of the cortex, DMN areas tend to increase and decrease their activity levels in concert with one another during long periods of “rest” (i.e., spontaneous mental activity) as well as during periods of overt task performance (Greicius, Krasnow, Reiss, & Menon, 2003; Srivastava, Reiss, & Menon, 2004). The idea received further support from a cortical parcellation study based on intrinsic functional connectivity data from 1000 participants, where unsupervised clustering of cortical parcels showed that the DMN naturally emerges as one of several large-scale networks (Yeo et al., 2011). Seed-based functional connectivity analyses also showed that each DMN region is more tightly coupled with other portions of the DMN than with any other cortical areas during spontaneous mental activity (Yeo, Krienen, Sepulcre, Sabuncu, Lashkari, et al., 2011).

However, it is important to note that the DMN lacks a formal definition. While a broad topographical pattern is consistently found in resting-state functional connectivity studies, the borders of the different DMN regions can vary substantially depending on the parcellation parameters (Yeo et al., 2011). Because each node encompasses multiple functional areas (Glasser, Coalson, Robinson, Hacker, Harwell et al., 2016), some of these areas can be included or excluded depending on the particularities of the study. Moreover, some DMN nodes are more reliably activated than others across task-based studies.

Based on intrinsic functional connectivity, Andrews-Hanna and colleagues (Andrews-Hanna, Reidler, Huang, & Buckner, 2010; Andrews-Hanna, Saxe, & Yarkoni, 2014) proposed that the DMN consists of three functional subsystems: the “core”, including the medial parietal node, the rostral medial prefrontal cortex, and the middle portion of the AG (roughly corresponding to area PGs); the “medial temporal lobe” subsystem (which also includes the RSC and the posterior portion of the AG); and the “dorsomedial” subsystem (which also includes the IFG, the anterior portion of the AG, and the lateral temporal cortex). They suggested that the medial temporal lobe subsystem is more directly involved in constructing a mental scene based on memory, while the dorsomedial subsystem is more involved in processing self-related information and inferring the mental states of others. Wen, Mitchell, & Duncan (2020) found mixed evidence for this functional fractionation of the DMN, concluding that “there is evidence of a largely integrated whole, with similar whole-brain activity maps for multiple contrasts, and some response to every contrast in each of the proposed subsystems,” and that “within subsystems and even within individual ROIs, different types of cognitive operation produce different local patterns of activity.” A very different fractionation scheme has been proposed by Braga & Buckner (2017) and DiNicola, Braga, & Buckner (2020), who found evidence for two functional subnetworks similarly distributed across most of the canonical DMN regions by conducting detailed functional connectivity studies of individual participants. The main distinction between them

was that one subnetwork showed functional connectivity with the hippocampus, while the other did not. This fractionation into interdigitated subnetworks is somewhat inconsistent with the scheme proposed by Andrews-Hanna and colleagues. At this point, what can be concluded from these studies is that the different areas that make up the DMN likely perform distinct computations while working together to enable a more global function.

2. DMN areas are engaged during semantic tasks

The earliest evidence that DMN regions are involved in semantic cognition came from Binder, Frost, Hammeke, Bellgowan, Rao, et al. (1999). The authors hypothesized that task-induced deactivation resulted from interruption of what James (James, 1890) called the “stream of consciousness”, which they conceived of as consisting mainly of “semantic retrieval and knowledge utilization” processes. According to this hypothesis, task-induced deactivation should depend on the semantic demands of the task. That is, if task-induced deactivation occurs because of interruption of semantic retrieval processes, then a task that engages semantic retrieval should produce less deactivation than a task that does not. The authors compared BOLD signals obtained during four states: a “resting” condition with no overt task (Rest), a non-semantic perceptual discrimination task involving auditory tones (Tones), a non-semantic phoneme discrimination task (Phonetic), and a semantic retrieval task involving feature judgments on auditory words (Semantic). In agreement with their hypothesis, the contrast Tones - Rest produced “deactivation” in most of the DMN areas in the left hemisphere, whereas the Semantic - Rest contrast produced only a small deactivation in the right pCinG. The contrasts Semantic - Tones and Semantic - Phonetic both showed positive activations in several DMN regions.

These results appeared to conflict with the findings of Shulman et al. (1997), who reported deactivation (relative to rest) in two language-related tasks. This discrepancy likely arises from the relatively “shallow” semantic processing required by the tasks studied by Shulman et al., which consisted of reading aloud and verb generation in response to presented nouns. Neither of these tasks require extensive retrieval of semantic features, unlike the semantic judgment tasks in Binder et al. (see Binder, 2012, for further discussion). Relative to reading aloud, verb generation produces activation in several regions associated with cognitive control, including the left dorsolateral prefrontal cortex and anterior cingulate gyrus (Raichle, Fiez, Videen, MacLeod, Pardo, et al., 1994), but no involvement of typical DMN regions.

Binder, Swanson, Hammeke, & Sabsevitz (2008) replicated the semantic task dependency effect in another fMRI experiment using the same tasks as in their earlier study (Figure 2A). They showed furthermore that the network of brain regions activated in the Semantic > Phonetic contrast, which is designed to highlight semantic-related processes, is virtually identical to the network showing task-induced deactivation in the Tones-Rest contrast (Figure 2B). Wirth, Jann, Dierks, Federspiel, Wiest, et al. (2011) tested the same hypothesis in a study where, in different blocks, participants made semantic, phonological, and perceptual decisions on the same words. They were instructed to respond “yes” or “no” to the questions “does the word refer to living objects”, “does the word have two syllables”, and “does the word contain perceptually identical letters”. Task blocks were alternated with

passive visual fixation blocks. The authors used independent components analysis (ICA) to identify the DMN in each subject. As a whole, the DMN was deactivated relative to rest in all three task conditions, but it was significantly less deactivated in the semantic condition than in the other two, and this result was also found separately for left DMPFC, rostral anterior cingulate, AG, RSC, MTG, and temporal pole. Other authors have also replicated the overlap between regions showing deactivation during non-semantic tasks and activation during semantic tasks (Seghier & Price, 2012; Shapira-Lichter, Oren, Jacob, Gruberger, & Hendler, 2013).

The large meta-analysis of semantic imaging studies by Binder, Desai, Graves, & Conant (2009) provides further evidence supporting a semantic or conceptual function for the DMN (Figure 1B). This analysis applied the activation likelihood estimation (ALE) approach to results from 87 published imaging contrasts (691 activation peaks) targeting general semantic processing. Included contrasts had to use verbal stimuli (spoken or written language) and have adequate controls for non-semantic (i.e., orthographic, phonological, and domain-general cognitive control) processes. The network identified with these methods closely resembles the DMN and includes the AG, pCing and precuneus, DMPFC and VMPFC, lateral temporal cortex, PHG, and IFG.

Three other studies designed to examine the relationship between the DMN and the set of areas involved in semantic processing have produced inconclusive results. Humphreys, Hoffman, Visser, Binney, & Lambon Ralph (2015) combined data sets from four different studies, including seven different tasks, to examine the similarities and differences between these networks. The DMN was defined by the contrast “rest > task”, while the language network was defined as “semantic task > rest”. Taking passive rest as the common baseline across different semantic and non-semantic tasks, the authors concluded that only a small portion of the DMN located in the anterior temporal lobe was involved in semantic processing. However, given the evidence reviewed above that participants do engage in semantic processing during rest – as acknowledged by the authors themselves – contrasts against rest are ill-suited to assess whether a region is involved in semantic cognition (for other compelling arguments against using passive rest as a baseline in cognitive neuroimaging, see Morcom & Fletcher (2007). Furthermore, by including “rest > semantic task” in the definition of the DMN, the study design was strongly biased against finding similarities between the two networks. The fact that both the DMN and the semantic network, as defined in the study, included primary sensory areas further indicates inconsistencies with that approach.

Jackson, Cloutman, & Lambon Ralph (2019) examined both resting-state connectivity and task-based fMRI data to investigate the same question. The DMN was defined as the set of areas whose time course correlated with a seed ROI in the rostral medial prefrontal cortex, while the semantic network was defined as the areas correlating with a seemingly arbitrarily chosen seed in the ventral anterior temporal lobe. Functional connectivity analyses revealed “highly overlapping seed-based results” between the two networks. The authors then performed ICA on the resting-state data set and identified the components (i.e., maps) that were most similar to the seed-defined DMN or to the seed-defined semantic network. The identified components were then used as templates in an ICA analysis of the task data aimed

at identifying independent components associated with semantic processing or the DMN. By examining how these last semantic and DMN components related to the contrast map “semantic task > non-semantic task”, they determined that two components were related to semantic processing alone, one component was related to the DMN alone, and one component was related to both, suggesting that some portions of the DMN are involved in semantics but others are not. Importantly, these results are very likely to be specific to the (somewhat arbitrary) particularities of the analyses, including the location of the network-defining seeds and the experimental conditions used in the univariate contrast (i.e., similarity judgment on words versus similarity judgment on letter strings, not accounting for differences in phonological processing or eye movements). Furthermore, the ICA approach used to identify semantic and DMN components is a signal separation technique designed to find differences between the two networks, as well as fractionations within each network, and it is unclear how these model-free fractionations relate to cognitive processes.

Using functional connectivity analysis, Braga, DiNicola, Becker, & Buckner (2020) reported identifying a “language network” that is distinct from and adjacent to the DMN. However, this study did not actually investigate whether the DMN is involved in semantic processing. The “language network” was defined *a priori* as the set of regions that (1) are more strongly activated for sentences than for nonword lists and that (2) have strong connectivity with an activation peak located in the posterior middle frontal gyrus, but that (3) cannot be considered to be part of the DMN or the frontoparietal control network. Not only is this a surprisingly arbitrary definition, but it requires the proposed language network to be distinct from the DMN, with no clear justification. That is, the finding that the language network is distinct from the DMN was predetermined by the analysis procedures.

All task-based fMRI activation studies reviewed above are based on “cognitive subtraction” (i.e., the difference in signal intensity between two experimental conditions), which has well-known limitations (Friston, Price, Fletcher, Moore, Frackowiak, & Dolan, 1996; McKiernan, D’Angelo, Kaufman, & Binder, 2006; Sartori & Umiltà, 2000). The rationale for the approach is that, if the two conditions are similar in all respects except for the degree to which they engage the process of interest, the resulting activation can be taken as an index of how much that region contributes to the process. The validity of this approach relies on assumptions that are unlikely to be true in many circumstances (e.g., pure insertion, equal automaticity), but even when these assumptions are reasonably met, detailed task analysis is required to characterize the processing differences between the two conditions. In other words, an fMRI contrast can only be interpreted in terms of the known differences between conditions.

Fernandino, Binder, Desai, Pendl, Humphries, et al. (2016) dealt with this issue by parametrically manipulating semantic features of visually presented nouns in a single task, which provides a more specific characterization of the process driving the BOLD signal. Participants performed a concrete/abstract decision on a set of 900 nouns spanning a wide range of imageability ratings and various semantic categories, such as manipulable objects, nonmanipulable objects, animals, emotions, professions, time spans, and events. A separate group of participants rated each noun according to the importance of five features, namely color, shape, manipulability, motion and sound. The ratings were used

as predictors in the GLM analysis on the fMRI data, resulting in five whole-brain maps of where the BOLD signal correlated with each rating. Activation related to the five semantic features were found across widespread cortical areas, including all cerebral lobes. They included association areas in the parietal, temporal and frontal lobes, as well as sensory-motor areas in the postcentral gyrus and lateral occipital cortex (for shape), the middle portion of the superior temporal sulcus (for sound), the posterior MTG and anterior SMG (for manipulability), and visual areas V4/V8 (for color), in agreement with “grounded” or “embodied” views of concept representation. The key idea of embodied theories of semantics is that the same cognitive/neural representations that underlie perceptual and motor processes also support semantic cognition, with no need for a distinct, symbolic representational system (Barsalou, 1999, 2008; Binder and Desai, 2011; Glenberg, 1997; Pulvermüller, 1999; for reviews of the evidence supporting this idea, see Borghesani and Piazza, 2017; Kiefer and Pulvermüller, 2012; Kuhnke, Beaupain, Arola, Kiefer, & Hartwigsen, 2023; Meteyard, Cuadrado, Bahrami, & Vigliocco, 2012). Importantly, only in typical DMN regions (AG, pCinG/precuneus, RSC, and DMPFC) did activity correlate with all five feature ratings, suggesting that these regions may play a role in integrating information originating from various sensory-motor cortical areas during semantic cognition. Similar evidence of such a role for the AG was found by Bonner, Peelle, Cook, & Grossman (2013) using a lexical decision task, which does not require explicit semantic processing but is thought to activate lexical semantic representations (i.e., word meanings).

3. DMN areas encode experiential information related to individual lexical concepts

Compared to univariate approaches, multivoxel pattern analysis (MVPA) methods provide a more specific assessment of the information content encoded in a region of the brain during a task (Kriegeskorte, Mur, & Bandettini, 2008). Studies using MVPA approaches confirm and extend the findings discussed above, further implicating the DMN in semantic processing (e.g., Clarke & Tyler, 2014; Devereux, Clarke, Marouchos, & Tyler, 2013; Fairhall & Caramazza, 2013; Fernandino, Humphries, Conant, Seidenberg, & Binder, 2016; Neyens, Bruffaerts, Liuzzi, Kalfas, Peeters, et al., 2017; Viganò, Borghesani, & Piazza, 2021). In an encoding-model MVPA analysis of the same data set from Fernandino, Binder, et al. (2016a), Fernandino, Humphries, et al. (2016b) provided further support for the idea that semantic representations in association cortical areas, including the DMN, encode sensory-motor information from multiple modalities during semantic word processing. The analysis evaluated whether the multivoxel activation patterns elicited by each word in these areas could be predicted based exclusively on five sensory-motor feature ratings of the word’s referent and on how each voxel responded to those feature ratings during presentation of other words. Prediction accuracy was significantly higher than chance level in the ROI defined by the semantic meta-analysis of Binder et al. (2009), but not in a control ROI encompassing the areas involved in word form analysis. Conversely, a set of five word-form attributes (number of letters, number of syllables, orthographic neighborhood density, phonological neighborhood density, and bigram frequency) successfully predicted

the activation patterns elicited by the word stimuli in the word-form ROI, but not in the ROI that included the DMN areas.

This result was confirmed and extended in a subsequent study using representational similarity analysis (RSA) of two independent data sets (Fernandino, Tong, Conant, Humphries, & Binder, 2022). In the “model-based” RSA approach, a matrix of all possible pair-wise stimulus similarity values is constructed using a similarity metric derived from a quantitative model of semantic content. RSA is performed by correlating the model-derived pairwise similarity matrix with a pairwise similarity matrix derived from the multivoxel activation patterns produced by each stimulus in a given ROI. Fernandino et al. used this technique to compare seven different models of semantic representation in terms of how well they predicted the neural similarity matrix for hundreds of English nouns. They found that a model consisting of eight sensory-motor features of word meaning predicts the similarity structure of word-specific fMRI activation patterns, even after partialling out the similarity patterns predicted by models based on word co-occurrence statistics (i.e., distributional information), taxonomic structure, and semantic features spontaneously produced by research participants. A larger set of 48 experiential features (including sensory-motor, spatial, temporal, causal, and affective features) performed even better; in fact, it accounted for all the variance explained by any of the distributional, taxonomic, and feature-production models evaluated. These results indicate that the semantic information encoded in DMN regions is more directly related to the sensory-motor-affective structure of phenomenological experience (i.e., experiential information) than to other kinds of information. Similar results were obtained when the analyses were performed in each DMN region separately, indicating that all regions encode experiential information related to word meaning.

Finally, recent results using searchlight RSA to map brain regions that encode single-word conceptual content revealed a network that encompasses the DMN. Tong, Binder, Humphries, Mazurchuk, Conant, & Fernandino (2022) performed a large-scale searchlight RSA study combining data from 64 participants and BOLD responses to 522 nouns covering a wide range of object and event concepts. Participants performed a general familiarity judgment on each word. Tong et al. used the model developed by Binder et al. (2016) consisting of importance ratings for 65 experiential features. The searchlight method used by Tong et al. applies the same RSA approach repeatedly to small ROIs centered on each vertex of the cortical surface, producing a map of RSA correlation values. As shown in Figure 1C, a network of brain regions showed patterns of neural response similarity across stimuli that matched the similarity patterns predicted by the experiential semantic model. This network includes all typical DMN areas, encompassing midline parietal and prefrontal cortices, AG, lateral temporal cortex, SFG, pMFG, and IFG. Partialling out the similarity structure predicted by distributional or taxonomic models did not change the results, indicating that they were indeed driven by experiential similarity between concepts.

The RSA map also includes other association cortical areas outside the DMN, such as the orbitofrontal cortex, inferior frontal sulcus and dorsal IFG, the inferior precentral sulcus, a portion of the IPS, anterior SMG, and the posterior end of the MTG and adjacent inferior temporal sulcus. Most of these, including the dorsolateral frontal cortex, IPS, and

pMTG, have been proposed to provide cognitive “control” processes during semantic tasks, i.e., processes that direct attention and selection mechanisms based on task requirements (Jackson, 2021; Lambon-Ralph, Jefferies, Patterson, & Rogers, 2017; Thompson-Schill, D’Esposito, Aguirre, & Farah, 1997). The finding that these areas encode word-specific semantic information during the task may reflect tuning, at a relatively fine spatial scale, to particular sensory-motor and affective features stored elsewhere, which would be expected if they are involved in controlled retrieval or short-term maintenance of semantic representations that combine information originating from a wide variety of cortical areas. Thus, it is possible that the detection of lexical semantic information in these areas is a consequence of the explicit semantic judgment task used by Tong et al., which is thought to engage control processes in addition to the experiential representations encoded in the DMN. This could explain why, unlike the DMN, they are not engaged during “resting” and passive states, which require little or no selection or attention processes and are typically experienced as “mind wandering”, “daydreaming”, and “stream of consciousness”. On the other hand, the pMTG, anterior SMG, anterior IPS, and ventral precentral sulcus have all been implicated in motor cognition, particularly object-directed actions (Caspers, Zilles, Laird, & Eickhoff, 2010; Gallivan, McLean, Valyear, & Culham, 2013; Grosbras, Beaton, & Eickhoff, 2012), and may be involved in representing action-related features of word meaning (about one-quarter of the words in the study referred to manipulable objects).

It is critical to note that the fMRI studies described above mainly focused on BOLD responses to single words (i.e., lexical concepts), and the analyses were constructed in ways that combine responses across a wide range of semantic categories. Together, these various studies indicate that the typical DMN areas are strongly involved in processing word meaning, and that other association areas outside of the DMN also contribute to semantic cognition at the single word level. Finally, and particularly important for our goal in this article, these findings indicate that all of the areas identified encode information about the relative importance of various sensory-motor and interoceptive neuro-cognitive systems to the meaning of a word, beyond what would be expected based on taxonomic structure or statistical patterns of word co-occurrence.

4. DMN areas encode the overall meanings of sentences and narratives

Several studies indicate that the DMN also represents compositional semantic information, that is, the overall meaning of phrases, sentences, and narratives. In their 2008 ALE meta-analysis focused on text comprehension, Ferstl and colleagues examined 10 studies contrasting coherent, comprehensible text with narratively incoherent text such as word lists or unrelated sentences (Ferstl, Neumann, Bogler, & Von Cramon, 2008). The results implicated much of the DMN, including MPFC, pCinG, precuneus, bilateral superior temporal sulcus and MTG, and left temporoparietal junction. Extensive subsequent work, including studies using semantic decoding models to predict activation patterns during narrative comprehension, has replicated these findings and explicitly emphasized a role for the DMN, including the angular gyrus as well as the regions identified by Ferstl et al., in processing sentence- and discourse-level semantics (Acunzo, Low, & Fairhall, 2022; Anderson, Kiela, Binder, Fernandino, Humphries, et al., 2021; Chen, Leong, Honey, Yong, Norman, & Hasson, 2017; Dehghani, Boghrati, Man, Hoover, Gimbel, et al., 2017;

Lerner, Honey, Silbert, & Hasson, 2011; Tylén, Christensen, Roepstorff, Lund, Østergaard, & Donald, 2015; Xu, Kemeny, Park, Frattali, & Braun, 2005; Yang, Li, Lin, Zhang, Wang, Zhang, et al., 2019; Yarkoni, Speer, & Zacks, 2008). Interestingly, the study by Xu et al. provided robust evidence that, in a reading task, core DMN regions are more engaged when the word sequences constitute meaningful sentences than when they are randomly ordered, and even more engaged when the sentences form meaningful narratives. Moreover, their activation was significantly stronger toward the end of the narratives (the conclusion segment) than during the initial portion (the setting segment). Further evidence was provided by Simony, Honey, Chen, Lositsky, Yeshurun, et al. (2016), who found that the inter-subject functional correlation (ISFC) in DMN areas was stronger when listening to a coherent story than when listening to a version of the same story in which paragraph order was scrambled, and not significantly different from zero when word order was scrambled (Figure 3A).

Together, these studies suggest that DMN areas not only encode information about individual word meanings but also encode internal models of novel situations and events built from meaningful word combinations during sentence and narrative comprehension. On the basis of these observations, and consistent with our view, several authors have proposed that a major function of the DMN is to construct internal models of events, whether directly experienced or imagined, by integrating information provided by more specialized neural systems (e.g., Baldassano, Chen, Zadbood, Pillow, Hasson, & Norman, 2017; Chen, Leong, Honey, Yong, Norman, & Hasson, 2017; Dehghani et al., 2017; Tylén et al., 2015; Zadbood, Chen, Leong, Norman, & Hasson, 2017). This process is required not only for understanding language but for many other cognitive tasks. In the following, we briefly review evidence on the involvement of the DMN in such tasks.

5. The DMN integrates experiential representations retrieved from memory

In trying to understand the functional role played by DMN areas in semantic processing, we can find clues in studies that showed involvement of those areas in tasks that did not explicitly manipulate semantic factors. Perhaps most prominent in the neuroimaging literature are studies investigating episodic memory retrieval. In the remember/know paradigm, participants decide whether each item in a list was included in a set of previously studied items (typically words or pictures) or is a new item. If they decide the item was in the original set (“old”), they must then indicate whether they clearly remember the experience of studying that item (“remember”) or are instead making that decision based merely on a sense of familiarity (“know”). Several neuroimaging studies have shown that, relative to correctly identified “new” items, correctly identified “old” items activate several DMN areas, including AG, pCingG, precuneus, RSC, VMPFC, and DMPFC (for reviews, see Hassabis & Maguire, 2007; Renoult, Irish, Moscovitch, & Rugg, 2019; Wagner, Shannon, Kahn, & Buckner, 2005). These regions also show stronger activity during trials that elicit detailed recollection of the event (Remember > Know). Some of these areas were also highlighted in a recent fMRI study that compared three episodic memory tasks involving different encoding strategies – sequential, spatial, and autobiographical (Flanagin, Klinkowski, Brodt, Graetsch, Roselli, et al., 2022). Correct item recall was associated with precuneus activation in all tasks, with RSC activation in both spatial and autobiographical conditions, and with PHG activation only in the autobiographical condition. Furthermore,

MVPA studies employing a variety of tasks and types of stimuli indicate that neural activity in DMN areas – particularly the AG and the precuneus/pCinG – does not merely track successful episodic recollection but encodes specific information about what is being remembered (Bainbridge & Baker, 2022; Chen et al., 2017; Rissman, Chow, Reggente, & Wagner, 2016; Zadbood et al., 2017). In the Chen et al. study, for example, participants watched a 50-minute long audiovisual movie and then were asked to recount the sequence of events verbally, in their own words and with no cues, while undergoing fMRI. Searchlight multivoxel pattern analysis identified a set of multimodal areas where activation patterns during recollection matched the activation patterns recorded during movie-watching on a scene-by-scene basis, within and between participants. These effects were strongest in the precuneus, PCinG, RSC, AG, posterior PHG, and lateral temporal cortex. Zadbood et al. (2017) replicated and extended these findings, showing that, in participants who did not watch the movie but listened to a verbal recollection of it, activation patterns in those areas during listening matched those recorded from different participants during movie watching and recollection. This correspondence indicates that those DMN areas represent the semantic content extracted from verbal narratives using a similar code to that used to represent experiential content during movie watching and episodic recollection (Figure 3B). Although the picture emerging from brain lesion studies is less clear, several reports have indicated the presence of episodic recall deficits in patients with posterior parietal cortex damage (see Berryhill, 2012, for a review).

Prospection – the mental generation of possible future situations – is another field of research that has implicated cortical areas included in the DMN. Tasks that require people to imagine future events typically activate medial prefrontal and medial parietal areas as well as the AG when compared to a variety of high- and low-level control conditions (Schacter, Addis, & Buckner, 2007; Spreng, Mar, & Kim, 2009). Several other studies have examined the neural correlates of prospective memory, that is, the ability to plan an action to be executed at a particular point in the future and to retrieve that plan from memory when the time arrives (e.g., remembering to take a medication at a certain time; see Burgess, Gonen-Yaacovi, & Volle, 2011, for a review). Prospective memory includes a prospective component, which tracks cues indicating that the moment to execute the action has arrived, and a retrospective component, which retrieves the planned action from long-term memory. Tasks designed to manipulate prospective memory also activate the medial prefrontal cortex, the precuneus/pCinG, and the AG.

As shown in Figure 1D, these same regions are also activated during mentalizing, that is, when participants make inferences about the mental states of other people (Andrews-Hanna et al., 2014; Spreng et al., 2009). In a “false belief” task, for example, participants are presented with a short narrative in which a character is expected to form a belief that the participant knows to be false or incorrect, thus requiring them to adopt a perspective different from their own to predict the character’s behavior. “False belief” stories and tasks involving attribution of intentions show activation in the AG, anterior MTG, DMPFC, and precuneus/pCinG/RSC when contrasted against stories that require inferences about physical causality without reference to mental states (e.g., Brunet, Sarfati, Hardy-Baylé, & Decety, 2000; Overwalle, 2009; Perner, Aichhorn, Kronbichler, Staffen, & Ladurner, 2006; Saxe & Kanwisher, 2003). A large quantitative meta-analysis of mentalizing studies by

Molenberghs, Johnson, Henry, & Mattingley (2016) found that the ventral aspect of the AG (roughly corresponding to area PGI) and the DMPFC were the most consistently activated regions across tasks and stimulus modalities. While some authors have interpreted these results in terms of mentalizing-specific cognitive processes, it seems to us that mentalizing-related task conditions typically require more complex inferential chains, based on a wider variety of semantic dimensions, than the control conditions used in these studies. Therefore, we propose that these activations reflect a difference in the complexity of the mental models built in the two conditions rather than the recruitment of computations or representations that are unique to mentalizing. This idea is supported by the results of Spreng and Grady (2010), who used spatio-temporal partial least squares to show that, in the same participant sample, autobiographical memory, prospection, and mentalizing share a common pattern of neural activity, with indistinguishable activation levels in AG, pCinG, RSC, DMPFC, VMPFC, IFG, and pMFG. Only in the posterior superior temporal sulcus and ventral supramarginal gyrus – areas outside the canonical DMN – was activation stronger for mentalizing than for the other conditions.

In addition to the tasks discussed above, activity in the DMN has also been prominently linked to the occurrence of “task-unrelated thoughts” (Singer, 1993), also known as “mind-wandering” (Smallwood & Schooler, 2006) or the “stream of consciousness” (James, 1890; McKiernan et al., 2006). McKiernan and colleagues showed that, even during performance of explicit tasks, the number of “task-unrelated thoughts” reported by participants during on-line probes was negatively correlated with task difficulty and positively correlated with activity in DMN regions (McKiernan, Kaufman, Kucera-Thompson, & Binder, 2003; McKiernan et al., 2006), findings subsequently replicated in other studies (Andrews-Hanna, Reidler, Huang, et al., 2010; Christoff, Gordon, Smallwood, Smith, & Schooler, 2009; Mason, Norton, Horn, Wegner, Grafton, et al., 2007). The information content of these phenomena has been studied in some detail, typically by eliciting subjective reports from study participants at random intervals during daily activities or experimental tasks (Antrobus, 1968; Giambra, 1995; Pope & Singer, 1976). Participants report a range of internally generated content, including verbal and nonverbal “images”, personal past events, general world knowledge, and future scenarios. We view “mind-wandering”, however, as a general term encompassing all forms of self-directed retrieval and manipulation of information stored in long-term memory, including semantic, episodic, and autobiographical memory retrieval, prospection and action planning, sensory and emotional imagery, and social cognition, rather than as a separate process somehow distinct from these.

Together, the studies reviewed above indicate that DMN areas are specifically involved in tasks that require the activation of sensory-motor and affective memory traces and their integration into internal models of scenes or events. Rather than attending to the current external environment or performing demanding perceptual analyses of the sensory input stream, these tasks require that elementary memory traces of shapes, textures, sounds, motion patterns, motor actions, colors, smells, and emotional experiences be selectively activated and integrated according to particular spatial, temporal, and causal relationships into a coherent model that can be used to guide action and/or predict future events. Such situation-model building is, in fact, a central aspect of semantic language processing (Glenberg, Meyer, & Lindem, 1987; Johnson-Laird, 1983; Kintsch, 1998; Zwaan

& Radvansky, 1998). But why are these areas, in particular, the ones most prominently involved in such tasks? We believe the answer lies in their unique positions in the connectivity structure of the cerebral cortex.

6. DMN areas are “untethered” from the sensory input stream

Anatomically, each of the DMN areas is maximally distant from primary sensory-motor areas in terms of geodesic distance across the cortical surface (Margulies et al., 2016). These areas are also the ones that mature latest in development (Buckner & Krienen, 2013; Hill, Inder, Neil, Dierker, Harwell, et al., 2010), with some of them (AG, lateral temporal cortex, pMFG, and IFG) showing the largest differences in organization between humans and macaques (Brodmann, 1909; Hill et al., 2010).

fMRI functional connectivity studies indicate that the DMN areas display a unique pattern of connectivity. They are most strongly connected with one another, showing weaker connectivity with other multimodal areas and the lowest degree of functional coupling to sensory-motor areas (Buckner et al., 2009; Margulies et al., 2016; Sepulcre, Sabuncu, Yeo, Liu, & Johnson, 2012). Step-wise connectivity analyses showed that these regions are the ones most distantly connected to any of the primary sensory and motor areas (Sepulcre et al., 2012), and Margulies et al. (2016) found that the principal gradient of functional connectivity across the entire cortex places the primary motor, somatosensory, auditory, and visual cortices together at one end, and seven regions of association cortex at the opposite end. These seven regions correspond precisely to the typical DMN areas (Figure 1E). These findings are supported by tract-tracing data from macaque and marmoset monkeys (Buckner & Margulies, 2019). Graph analyses of fMRI functional connectivity have indicated that some DMN areas, particularly the precuneus/pCinG, AG, and DMPFC, function as high-level connectivity hubs (Figure 1F), facilitating the integration of diverse types of information arising from various parts of the cortex (Buckner et al., 2009; Tomasi & Volkow, 2010).

A series of fMRI studies by Hasson and colleagues provides further evidence for the idea that the cortical areas making up the DMN are maximally independent of the activity taking place in early sensory-motor areas (Chen et al., 2017; Lerner et al., 2011; Simony et al., 2016). Using naturalistic stimuli lasting several minutes, such as movie clips and stories, they assessed the length of the “temporal receptive window” (TRW) across the cortex via inter-subject correlation of each voxel’s time course. The TRW of a voxel is the longest timescale in which information from previously presented stimuli affects that voxel’s response to new stimuli. They identified a cortical hierarchy of processing timescales extending from early sensory to association areas. TRWs were shortest in sensory areas and became gradually longer towards higher-order areas, peaking in the core DMN areas.

Discussing the evolution and development of the cortex, Buckner and Krienen (2013) point out that the functional organization of non-primary sensory and motor areas appears to be constrained (“tethered”) by the well-defined maps that characterize the corresponding primary areas. The degree of tethering of different areas decreases according to a spatial gradient that radiates away from these primary areas. They hypothesize that, in analogous

fashion, the development of multimodal areas is progressively less constrained by the organization of sensory-motor areas as their distance from these areas along the cortical sheet increases. According to this idea, the evidence reviewed above would suggest that the areas that constitute the DMN are the most “untethered” from ongoing experiential input.

We propose that such untethering allows DMN regions to selectively re-activate elementary experiential memory traces stored in multimodal and unimodal areas, integrating them into complex, composite representations of objects and events in the service of a goal, independent of ongoing experiential input. This proposal is supported by recent task-based fMRI studies examining changes in brain activation patterns as a function of the focus of attention. Regev, Simony, Lee, Tan, Chen, & Hasson (2019) used inter-subject correlation analysis to track how information was shared across cortical regions when participants were presented with two different verbal narratives simultaneously, one auditorily and the other in writing. Participants were instructed to attend to one of the stories and ignore the other. Activity related to the unattended story was confined to auditory areas (for spoken stories) or visual areas (for written stories), while activity related to the attended story was detected not only in those areas but also in multimodal cortical areas throughout the brain, particularly the AG and the medial parietal cortex. Cohen, Nakai, & Nishimoto (2022) presented subjects with audiovisual movie clips and instructed them to either attend to the movies or engage in one of three “internal” tasks, namely prospective mental imagery, episodic recall, or mental arithmetic. Attention to the movie clips resulted in stimulus-related activity (computed as the RSA noise ceiling across participants) being detected in all large-scale networks assessed, including the DMN. All three internal tasks, on the other hand, led to the DMN becoming less reflective of stimulus processing. Interestingly, there was an increase in functional connectivity between the DMN and the Visual, Attention, and Executive networks during prospective imagery and recall, whereas the arithmetic task, requiring only the retrieval of fixed relationships between decontextualized symbolic tokens, was marked by reductions in these same connections. These findings are consistent with the idea that the DMN is involved in goal-driven, top-down activation of experiential memory traces for the construction of composite representations. These multimodal representations can be created and modified independently from the sensory input stream – although the activation and integration of these memory traces can be guided by incoming information contained in the sensory input if that contributes to the individual’s goal (e.g., during movie watching and narrative comprehension).

7. A unifying theory of the DMN

We propose that the fundamental function of the DMN is to generate integrated models of particular states of the world, known in the cognitive psychology literature as “situation models” (Dijk and Kintsch, 1983) or “mental models” (Johnson-Laird, 1983). Situation models are analogical representations of scenes, events, or internal states built from information retrieved from long-term memory and maintained in an active state that allows them to be used to generate expectations, draw inferences, and guide action, or to be updated with additional information (for a review, see Zwaan and Radvansky, 1998). The term is most often used in the context of discourse comprehension to refer to “mental representations of the state of affairs described in a text rather than of the text itself” (Zwaan,

1999), but has also been used in reference to evolving mental models extracted from visual narratives (i.e., a sequence of images that tells a story; Cohn, 2019).

An essential aspect of the notion of situation model is that the representation that results from comprehending a sentence, paragraph, or visual narrative is built, for the most part, from experiential information previously stored in long-term memory and from inferences based on that information, rather than from information contained in the stimulus itself. The stimulus simply provides instructions on what information to retrieve and constraints on how to combine the different elements into a coherent model. Note that what makes a model “coherent” is the comprehender’s knowledge of how the retrieved elements can be combined given their experiential properties, possibly implemented as a constraint satisfaction network (Hopfield & Tank, 1985; Rutishauser, Slotine, & Douglas, 2018; see Glenberg, 1997, for the related idea of “mesh”). For example, the sentence “Sally knocked over the coffee cup on her desk and damaged her notebook” makes sense because we can easily combine the different elements into a realistic spatio-temporal situation in which the damage to the book could be predicted from the knocking over of the cup. Replacing the word “notebook” with the word “roof” renders the sentence puzzling because it is difficult to imagine a realistic scenario that can accommodate it. The original sentence does not specify how far from Sally the notebook is, but we fill in that information with a reasonable estimate based on our previous experiential knowledge of how coffee spills typically behave. This knowledge also leads us to assume that the notebook was sitting on the same desk as the cup, rather than on a different piece of furniture. That is, the overall configuration of the situation model is constrained not only by the experiential properties of the entities and actions named in the stimulus and the syntactic structure of the sentence, but also by the mutual constraints that the experiential elements retrieved from memory impose on each other.

It is clear that situation models are required for the performance of any of the tasks that engage the DMN, not only for discourse comprehension. Episodic recollection, prospective imagery, theory-of-mind reasoning, and movie watching all depend on the internal generation of spatio-temporal situation models, whereas “externally directed” tasks such as visual search, auditory tone discrimination, and n-back matching with meaningless stimuli do not, no matter how difficult they may be.

Perhaps less obvious is the fact that semantic processing of isolated words also requires multimodal situation models, that is, the selective retrieval and integration of different types of experiential information into a coherent whole. Understanding a concrete noun such as “hammer” or “tiger” requires that information about shape, size, weight, texture, color, affective responses, manipulation affordance, and motion patterns be simultaneously activated, integrated, and made available for the generation of inferences, expectations, and action plans. Furthermore, contextual information associated with individual objects is automatically retrieved from memory during visual object recognition and during semantic processing of object nouns (Bar, Aminoff, & Schacter, 2008; Bar & Aminoff, 2003; Bonner & Epstein, 2021; Turini & Vö, 2022). Event nouns (e.g., “party”, “conference”, “apology”) and verbs arguably make even stronger demands on situation model building, while more abstract words such as “democracy” require additional contextual information to guide the building of detailed situation models (although people might generate relatively detailed but

idiosyncratic situation models in response to such words presented in isolation; see Davis, Altmann, & Yee, 2020, for an extended discussion of this point). However, this theory would predict that the DMN is comparatively less engaged during single-word processing than during tasks that require more complex situation models, consistent with the studies reviewed above.

We propose that the DMN enables the building of situation models by encoding patterns of coactivation among neuron populations located in various unimodal and multimodal cortical areas (Damasio, 1989). According to this proposal, these neuron populations store detailed representations of particular features of experience and participate more directly in perception and action. Different objects, scenes, and events, characterized by unique patterns of simultaneous activation across these areas, would then be stored in compressed form in the DMN (Kaefer, Stella, McNaughton, & Battaglia, 2022; McClelland, McNaughton, & O'Reilly, 1995). Concept retrieval would begin with the activation of such coactivation patterns in the DMN, which would then activate the respective feature representations in lower-level areas. While the activation of these areas would be initiated by the DMN, the flow of information would be bi-directional, resulting in changes in the DMN's activation patterns in response to the combined input from those areas (Cowansage, Shuman, Dillingham, Chang, Golshani, et al., 2014; Wolfrum, Wolff, Lücke, & Malsburg, 2007). As we conceive it, a situation model is implemented as a dynamic, transcortical neuronal assembly involving much of the cerebral cortex, subcortical structures, and cerebellum, with different regions contributing different kinds of experiential information according to their respective roles during perception and action (Barsalou et al., 2003; Binder & Desai, 2011; Kemmerer, 2015; Kiefer & Pulvermüller, 2012). By entraining its neuronal activity to the activity of particular subpopulations of neurons in each area, the DMN may provide the means for these various neuronal groups to temporarily influence each other, so that the entire system can converge into a state of equilibrium (i.e., an attractor state) corresponding to a coherent situation model. The situation model would remain available for cognitive operations such as generating predictions, drawing inferences, or updating of the model itself – that is, it would remain active in working memory – for the duration of the attractor state.

In this view – which, at this point, remains speculative – the DMN itself represents semantic content at a level of representation in which information about various experiential modalities is integrated and compressed, resulting in a lack of experiential detail. This compressed, experientially impoverished content is what allows the DMN to modulate (via reciprocal connections) the appropriate lower-level representations stored in other areas, which, we propose, provide additional experiential information as needed to achieve the intended goal (for evidence of context-dependent recruitment of sensory-motor representations, see Kuhnke et al., 2020; Kuhnke, Kiefer, & Hartwigsen, 2023; Muraki, Doyle, Protzner & Pexman, 2023; van Dam, Brazil, Bekkering, & Rueschemeyer, 2014; van Dam, Dijk, Bekkering, & Rueschemeyer, 2011; van Dam, Rueschemeyer, Lindemann, & Bekkering, 2010; Wang, Gao, Smallwood, & Jefferies, 2021). The DMN's record of coactivation patterns, by itself, may not be sufficient to support a situation model that is distinct and detailed enough for the task at hand, and the additional information stored in lower-level areas may be essential. Thus, each DMN, lower multimodal, and unimodal area

probably encodes unique semantic information that contributes to the generation of situation models, although at different levels of integration and detail. In other words, we suggest that the DMN permanently stores semantic information in the form of patterns of coactivation among lower-level cell assemblies.

In addition to these long-term latent representations, the DMN would also temporarily represent novel patterns of coactivation, which could be guided by verbal input (e.g., during narrative comprehension) or by sensory-motor-affective information (e.g., when the individual experiences a novel situation). This novel pattern of coactivation could be consolidated into a long-term engram in the DMN if it were ecologically relevant or if it re-occurred repeatedly. If it were consolidated, it could be associated with a phonological word form, thus becoming a new lexical concept. Importantly, phonological word forms are represented in a distinct network (Binder et al., 2008; Graves, Desai, Humphries, Seidenberg, and Binder, 2010; Graves, Grabowski, Mehta, & Gupta, 2008, Yue and Martin, 2021; for a recent meta-analysis, see Hodgson, Lambon Ralph, and Jackson, 2021), and can be arbitrarily related to the experiential situation models to which they refer, as is the case with the vast majority of words (Ogden and Richards, 1923; Saussure, 1916/1983). While activation of the word form would typically induce activation of the associated mental model in the DMN and relevant downstream areas during language comprehension (and vice-versa during language production), activation of the mental model would not require activation of the word form, since we know each system can be damaged independently (Hodges and Patterson, 1996).

Although we propose a general role for the DMN in coordinating the activation and construction of situation models in working memory, that is not to say that the DMN is functionally homogeneous with regard to the information processed in different nodes. Although they are most strongly connected with one another, different DMN regions have distinct cytoarchitecture (Economo and Koskinas, 2007) and are differentially connected to other cortical and subcortical structures, which imply (at least partially) distinct functions. Clues to their individual contributions to situation model implementation are provided by functional neuroimaging and focal brain lesion studies. For example, the RSC and posterior PHG have been strongly implicated in tasks involving spatial navigation and topographical orientation. The PHG contains a functional area often called the “parahippocampal place area” (PPA), which has been repeatedly shown to respond preferentially to pictures of buildings, streets, and landscapes compared to pictures of individual objects such as artifacts, animals, faces, or body parts (Downing, Chan, Peelen, Dodds, & Kanwisher, 2006; Epstein & Kanwisher, 1998). As noted by Epstein (2008), the “PPA” label may be misleading, since this area responds more strongly to scenes (i.e., sets of objects in particular spatial configurations, including table-top scenes) than to buildings. The PHG and the RSC also activate during mental imagery of scenes and locations, and their response to pictures of individual objects can be modulated by the object’s navigational significance (Ino et al., 2002; Janzen & Turennout, 2004). Damage to these areas, which can occur following posterior cerebral artery strokes, causes persistent topographic disorientation, wayfinding deficits, and difficulty recognizing scenes, particularly with right hemisphere damage (Epstein, 2008; Maguire, 2001). Interestingly, there is also evidence that the RSC and the PHG are involved in coding contextual associations between objects (Bar, 2004; Bar et al.,

2008; Bar & Aminoff, 2003; Bonner & Epstein, 2021). For example, these areas are more engaged during the recognition of objects that are typically associated with specific spatial contexts (i.e., stoves are typically found in kitchens and cows are typically in pastures, but doors and cats are found in many different sorts of environments) and during visual processing of typical relative to atypical scenes. There is evidence, however, that these two regions play dissociable roles in spatial cognition (Persichetti & Dilks, 2019). The specific role of the RSC in these tasks – and in situation model building – might involve interfacing egocentric spatial experiences encoded in the parietal cortex with viewpoint-independent representations in the medial temporal lobe, enabling transformations between egocentric and allocentric coordinate systems (Alexander, Place, Starrett, Chrastil, & Nitz, 2022; Bicanski & Burgess, 2018), while the PHC may be more involved in encoding contextual or thematic associations between individual items. These areas may play an important role in activating and integrating schemas (i.e., stereotyped memory representations of common physical and social situations) into the developing situation model (see Barsalou, 2009, for the notion of “situated conceptualization”).

Compared to medial parietal regions, the VMPFC has been more associated with emotion regulation, decision making, evaluation of reward, and moral judgment. Compared to other DMN regions, the VMPFC has stronger connections with limbic structures such as the amygdala, ventral striatum, hypothalamus, lateral orbitofrontal cortex, and temporal pole (Barbas, 1995; Carmichael & Price, 1999; Yeo et al., 2011). Data from both rodents and humans indicate that this region regulates amygdala activity, mediating extinction of conditioned fear (Milad, Rauch, Pitman, and Quirk, 2006; Motzkin, Philippi, Wolf, Baskaya, & Koenigs, 2015; Phelps, Delgado, Nearing, & LeDoux, 2004). Activity in this area correlates with the subjective value of foods, nonfood consumables, social rewards, and money, suggesting that it may encode a “common currency” of value (Chib, Rangel, Shimojo, & O’Doherty, 2009; Levy & Glimcher, 2012), and this activity is decreased in pathological gamblers relative to healthy controls during tasks requiring inhibitory control (Potenza, Leung, Blumberg, Peterson, Fulbright, et al., 2003). Damage to this region is associated with changes in personality, emotional dysregulation, and deficits in complex decision making, resulting in more utilitarian moral judgments (Anderson, Barrash, Bechara, & Tranel, 2006; Koenigs, Young, Adolphs, Tranel, Cushman, et al., 2007). One study found that functional connectivity between posterior VMPFC, temporal-parietal junction, and middle temporal gyrus was correlated with individual differences in subjective value (Smith, Clithero, Boltuck, & Huettel, 2014). Because activity in these posterior regions accounted for unique variance in subjective value beyond that explained by the VMPFC, the authors interpreted these findings as indicating “not only that posterior VMPFC interacts with additional brain regions during valuation, but also that these additional regions carry information employed to construct the subjective value for social reward.” In a study investigating the role of the VMPFC in episodic simulation, participants were presented with arbitrary pairings of the names of a familiar person and a familiar location and imagined interacting with the named person in a location-specific manner (Benoit, Szpunar, & Schacter, 2014). VMPFC activity was modulated by the combined familiarity of the person and the location, and its connectivity to the PHC was modulated by place (but not person) familiarity, while connectivity with the DMPFC was mediated by person (but not

place) familiarity. Consistent with a role in representing affective quality, activity in the VMPFC also correlated with the rated pleasantness of the imagined episode. Thus, the main contribution of this region to situation model construction may relate to the integration of affective information with other model components.

As mentioned in the Introduction, each DMN node encompasses multiple functional areas (Glasser, Coalson, Robinson, Hacker, Harwell et al., 2016), and some of these ‘subnodes’ may be implicated in some task-based studies while not in others. In principle, this variability can be exploited to characterize the roles of different subnodes within a given node, although this may require a finer level of experimental control than what is typical in the literature, as well as techniques for co-registering different brains that account for intersubject variability in the mapping between functionally defined areas and patterns of cortical folding (Glasser et al., 2016; Haxby, Guntupalli, Nastase, & Feilong, 2020; Castello, Haxby, & Gobbini, 2021).

Many of the ideas discussed above have been proposed, in one form or another, by other authors. In particular, the idea that the DMN integrates different kinds of information to support internally generated representations has been espoused by several groups (e.g., Binder and Desai, 2012; Buckner et al., 2008; Hassabis and Maguire, 2007; Kaefer et al., 2022; Menon, 2023; Smallwood et al. 2021; Stawarczyk, Bezdek, & Zacks, 2021; Yeshurun & Hasson, 2021). Similarly, the notion that language meaning relies on internal simulations based on the same representational code as episodic memory, mental imagery, and perception has a long history (Barsalou, 1999, 2009; Damasio, 1989; Glenberg, 1997; Pulvermüller, 1999). The original contributions of this review include a critical survey of studies implicating the DMN in a wide range of semantic tasks, providing strong support for the idea that the DMN plays a central role in verbal and non-verbal semantic cognition, and the proposal of a more clearly specified mechanism of DMN function that accounts for these findings. In this proposal, DMN areas store information about patterns of coactivation among experiential representations in multimodal and unimodal areas, which allows it to flexibly reactivate those representations and integrate them via reciprocal connections into a transient, global activity pattern representing a situation model in working memory. While active, the situation model can be used to guide inferences and predictions or be combined with other representations retrieved from memory. While these ideas are consistent with many previous proposals, they provide a more fleshed out model of information processing in the DMN. The model not only accounts for the DMN’s involvement in the entire range of cognitive phenomena documented in the literature but also provides a mechanism for how concepts can refer to the external world and enable ecologically adaptive behavior (known as the “grounding problem”) while maintaining a level of independence from neural activity taking place in low-level sensory-motor areas. In this respect, the model is an elaboration of ideas advanced by grounded/embodied semantics and the notion of convergence-divergence zones (Tranel, Damasio, & Damasio, 1997).

Notably, this model diverges from the proposal that coherent, generalizable concepts are represented exclusively at a hub located in the anterior temporal lobe (Lambon Ralph, Jefferies, Patterson, & Rogers, 2017; Patterson, Nestor, & Rogers, 2007). Anatomically, the anterior temporal lobe (ATL) consists of all structures that are closer to the temporal pole

than to the temporal-occipital border. That includes the anterior portions of the superior, middle, and inferior temporal gyri, superior and inferior temporal sulci, fusiform gyrus, entorhinal cortex and hippocampus, plus the piriform cortex, temporal pole, and amygdala. Although it has been suggested that most of these structures contribute to semantic cognition in one way or another, to our knowledge there is no evidence that these structures constitute a functional unit. In fact, functional connectivity evidence indicates that they can be more strongly connected with areas outside of the temporal lobe than with each other (Murphy, Rueschemeyer, Watson, Karapanagiotidis, Smallwood, et al., 2017; Persichetti, Denning, Gotts, & Martin, 2021; Yeo et al., 2011). For example, the temporal pole is most strongly connected to the amygdala and to portions of the orbitofrontal cortex, and may be responsible for processing primarily interoceptive/affective information (Simmons, Reddish, Bellgowan, & Martin, 2010; Yeo et al., 2011); the anterior fusiform gyrus is part of the ventral visual stream and may encode detailed configurations of visual features that discriminate among unique objects within a category (Collins & Olson, 2014; Persichetti et al., 2021); the piriform cortex receives most of its input from the olfactory bulb, and thus processes mainly olfactory information (Gottfried, 2009; Torske, Koch, Eickhoff, & Freiherr, 2021); the entorhinal cortex appears to integrate information from multiple experiential modalities into spatial maps, encode arbitrary pair associations, and support the consolidation of declarative memories (Chao, Huston, & Silva, 2018); and the anterior superior temporal gyrus (STG), which is strongly connected to the posterior STG but not to anterior MTG or ITG, may encode primarily auditory (possibly phonological) information (Jackson, Hoffman, Pobric, & Lambon Ralph, 2016; Murphy et al., 2017). We propose that areas in the anterior MTG/STS that belong to the DMN (Figure 1) function as high-level hubs, as do areas in other canonical DMN nodes, and that these nodes work together to activate and integrate experiential memory traces into situation models. Therefore, we suggest that the label “anterior temporal lobe” (or “ATL”) be avoided when referring to any cortical area (or set of areas) believed to play a particular role in semantic cognition, and that more specific anatomical labels be employed instead.

The theory proposed here makes novel testable predictions. It predicts that the degree of involvement of the DMN in a task varies parametrically with the complexity of the situation model required. Complexity could be operationally defined in terms of the number of distinct experiential dimensions required to characterize the model, the level of detail or specificity required in each dimension, the number of entities and relationships represented in the model, or some combination of these. Internal models of individual, generic, uncontextualized sensory experiences (e.g., “yellow”) should be among the simplest. Models of events involving multiple well-defined intentional agents and complex spatial, temporal, and causal relations should be among the most complex and result in the strongest activation of the DMN.

Our proposal also predicts that, when words are presented in isolation, highly polysemous words (i.e., words that admit a wide variety of possible situation models depending on the context) will generate the least reproducible activation patterns in the DMN (i.e., the lowest RSA noise ceiling). Highly emotional words that may be considered abstract, such as “death” or “disgrace”, will generate more reproducible patterns. Unambiguous words for

stereotypical events with rich experiential content (e.g., “thunderstorm”) should generate the most reproducible activation patterns in the DMN.

Finally, if word meaning consists of a multimodal experiential situation model, as proposed here, we should expect that automatic semantic priming between individual words should be driven primarily by experiential similarity rather than by shared membership in natural categories or by measures of similarity derived from word co-occurrence patterns (i.e., distributional semantic models such as word2vec).

8. Conclusion

We conclude that DMN areas play an essential role in semantic cognition, from retrieval of word and object meaning to narrative comprehension and complex problem solving, and we propose that this role consists of constructing experiential situation models that are relatively uncoupled from current input to primary sensory-motor areas. They perform this function by coordinating the top-down activation of neuron subpopulations in various multimodal and unimodal cortical areas and allowing these segregated subpopulations to be transiently integrated into a transcortical neuronal assembly. DMN areas are uniquely positioned to perform this function due to their distance and functional independence from primary sensory-motor areas, as well as their connections with affective, motivational, and reward systems.

On this view, a lexical concept is not a fixed representation stored in any particular cortical region. Rather, it is a dynamic activation pattern in which neural activity across multiple multimodal, unimodal, and limbic cortical areas is coordinated and integrated via DMN connectivity hubs. The relative importance of each sensory-motor modality to the overall situation model during a particular instantiation of the concept depends on the contextual and motivational demands driving the construction of the model, and the degree of involvement of unimodal networks depends on the degree of experiential detail required. A corollary of this view is that no two instantiations of a lexical concept are identical, although they need to be sufficiently similar to lead to similar behavioral choices in similar situations. Furthermore, despite the functional dissociations documented in the literature, semantic, episodic, and procedural memory systems are considered to be interdependent and only partially dissociable. Situation models built during narrative comprehension are functionally and neurally akin to situation models retrieved as episodic memories, and procedural information such as sensory-motor skills can be activated and integrated into the situation model during semantic cognition.

This proposal integrates several important findings in the cognitive neuroscience of language and the DMN, including evidence that (1) cortical areas involved in perception and action are engaged during semantic language processing in a content-specific way; (2) the DMN is engaged by semantic language processing at word, sentence, and discourse levels; (3) the DMN is more engaged for coherent narratives than for disconnected sentences or unordered movie scenes, and more engaged for sentences than for lists of disconnected words; (4) each DMN area encodes information about word meaning that is more accurately predicted by experiential models than by taxonomic or distributional models; and (5) the DMN is

engaged by non-linguistic tasks that require situation model building, but not by tasks in which the main focus of attention is on incoming perceptual information. Building on these findings, we advance a neurocognitive mechanism for how concepts can refer to things in the external world and generate behaviorally adaptive predictions while maintaining an essential degree of independence from sensory input and motor output. This proposal makes specific empirical predictions that can be tested with currently available experimental techniques.

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Highlights

- There is strong evidence that the DMN is involved in semantic cognition
- DMN areas contribute to semantics at the word, sentence, and discourse levels
- The DMN enables the construction of embodied situation models
- Situation models are implemented as dynamic recurrent neural assemblies
- These assemblies include modality-specific, multimodal and DMN cortical areas

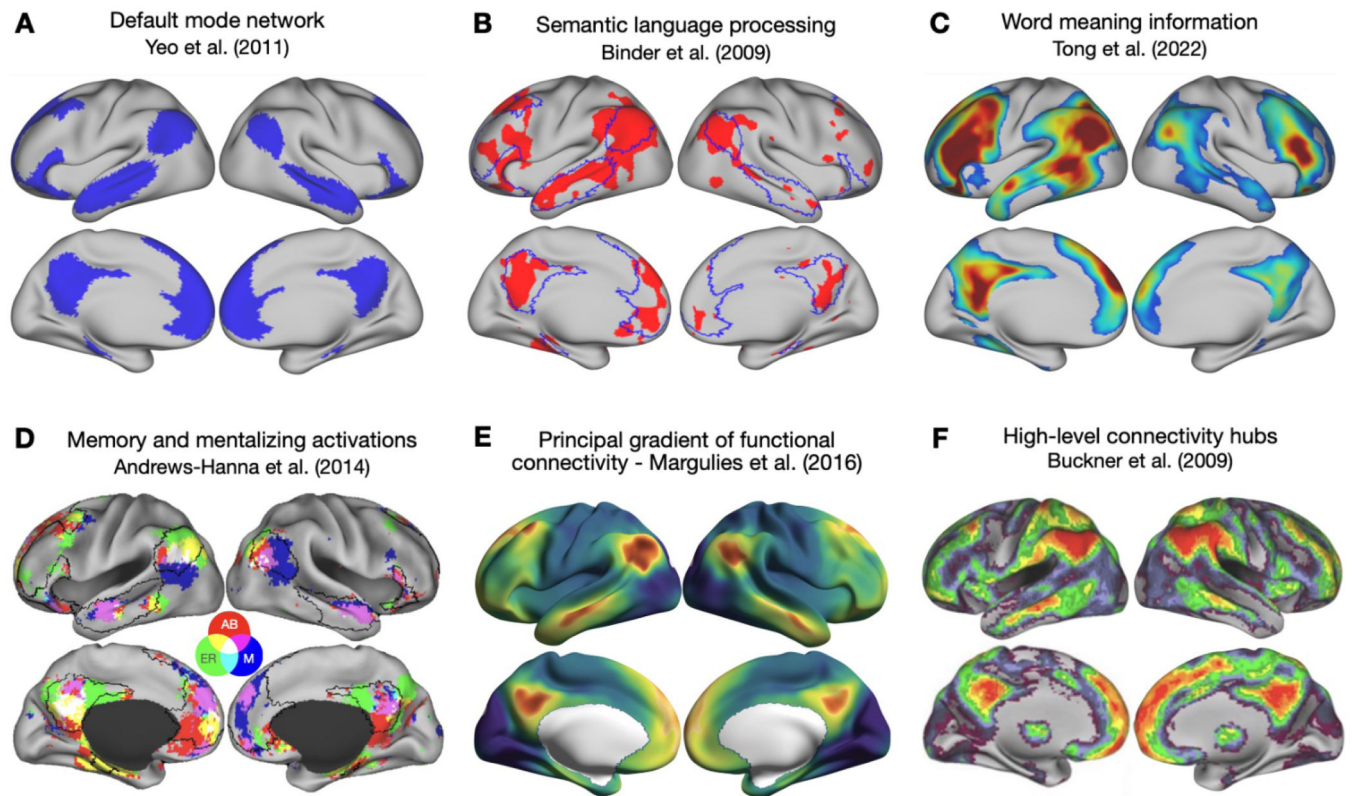
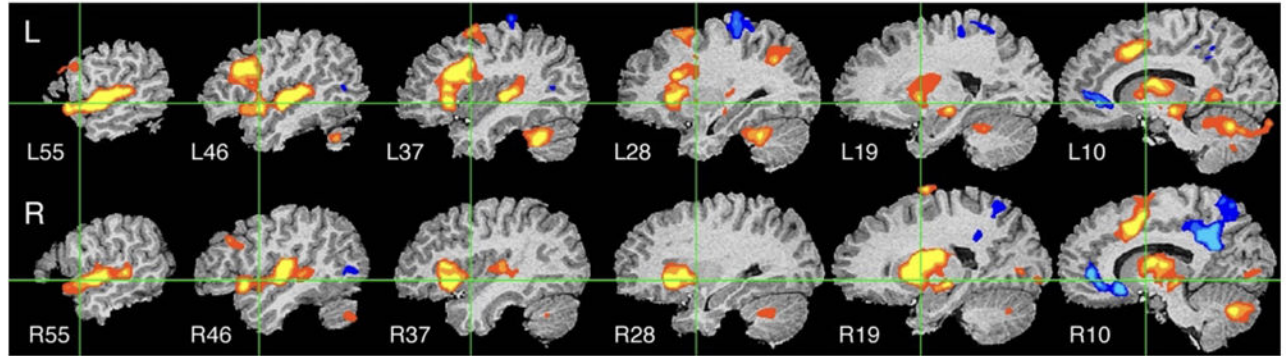
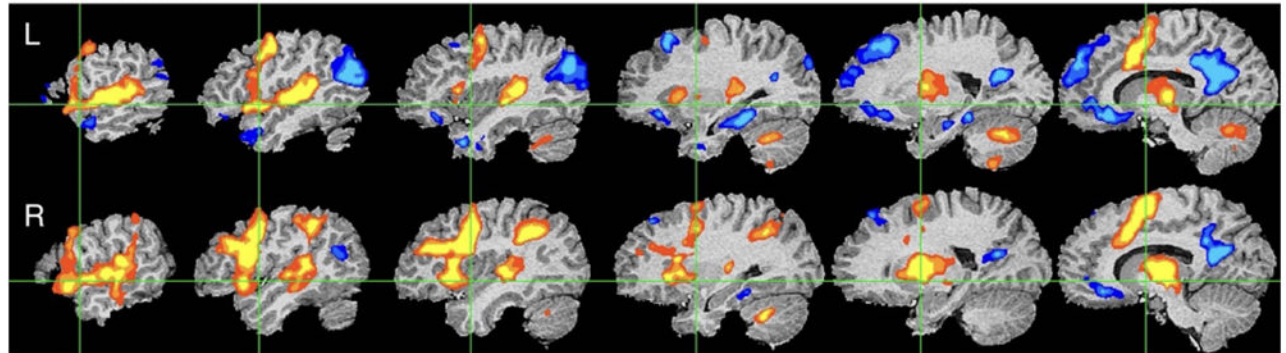
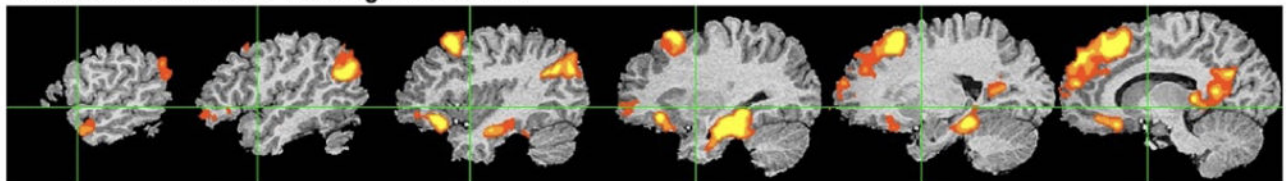
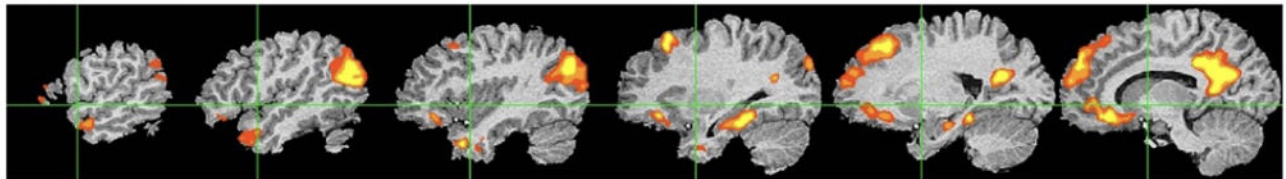


Figure 1.

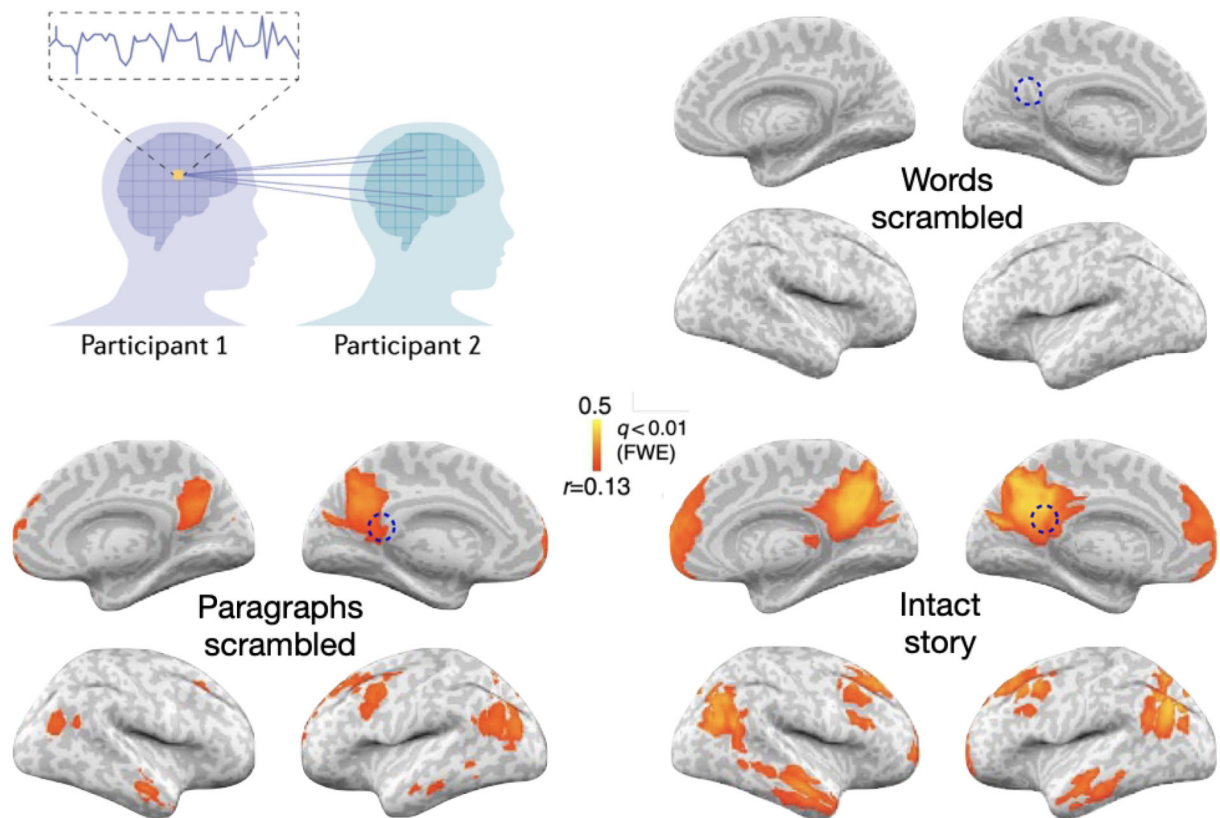
A: The canonical DMN as defined by resting-state functional connectivity parcellation (Yeo et al., 2011). B: Activation likelihood estimation meta-analysis of semantic language processing based on 86 studies (Binder et al., 2009). C: Mean RSA score for the experiential semantic model from Tong et al. (2022). D: Reverse inference maps from meta-analyses of episodic retrieval (ER), mentalizing (M), and autobiographical (AB) tasks (modified from Andrews-Hanna et al., 2014). E: Principal gradient of functional connectivity obtained by Margulies et al., 2016 (figure modified from Paquola et al., 2019). F: fMRI degree connectivity during rest (modified from Buckner et al., 2009; copyright 2009 Society for Neuroscience).

A Semantic Decision – Rest**Tone Decision – Rest****B Semantic Decision > Phonological Decision****Rest > Tone Decision****Figure 2.**

A. Differences in task-induced 'deactivation' by semantic (top) and non-semantic tone perception (bottom) tasks from Binder et al. (2008). Areas with lower BOLD signal during these overt tasks relative to a 'resting' state ('deactivation') are shown in blue colors. The angular gyrus, dorsomedial prefrontal cortex, parahippocampal gyrus, and left posterior cingulate gyrus show deactivation only during the non-semantic tone task. L = left, R = right. B. Similarity between the spatial pattern of the DMN, defined as regions showing relative deactivation during non-semantic tasks, and networks showing selective engagement by semantic (relative to matched control) tasks. Top: Semantic vs.

phonological task activation map. Bottom: Areas showing task-induced deactivation during a tone discrimination task.

A Inter-subject functional correlation using PCC as a seed



B Shared neural patterns across movie-viewing, recalling the movie, and listening to the movie recall.

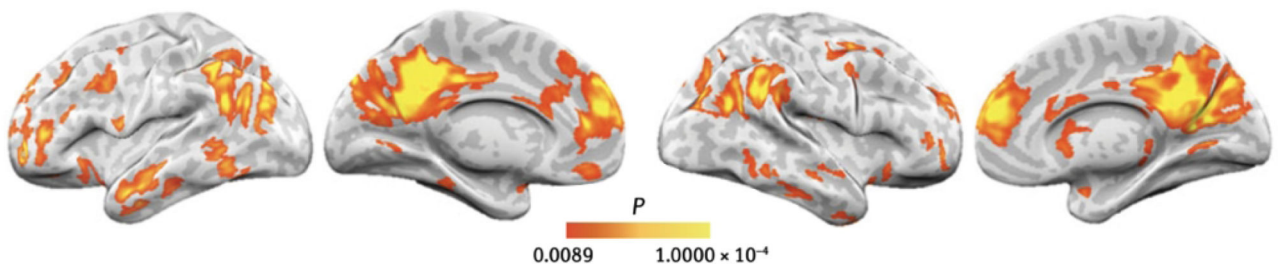


Figure 3.

A. Intersubject functional correlation between the pCinG seed (dashed circle) and the rest of the cortex across 18 participants in 3 experimental conditions (from Simony et al., 2016). B. Shared neural patterns across movie-viewing, recalling the movie, and listening to the movie recall from Zadbood et al. (2017). Figures modified from Yeshurun et al. (2021) and Simony et al. (2016).